

SMYRNA, TN, USA

WMO: 723273

Lat: 36.009N

Lon: 86.520W

Elev: 543

StdP: 14.41

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 13827

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.7	19.1	4.0	6.9	20.6	7.5	8.2	22.6	22.6	41.1	20.2	45.5	5.1	0	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	98.9	76.2	95.9	75.9	93.5	75.3	79.3	92.3	78.2	90.6	77.3	89.0	7.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.1	139.0	84.5	75.1	134.3	83.2	74.2	130.2	82.2	43.1	91.9	42.0	90.3	41.0	88.6	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.2	17.0	15.3	9.0	102.2	5.0	5.6	5.3	106.2	2.4	109.5	-0.4	112.6	-4.1	116.6		
			7.7	81.5	4.7	1.2	4.3	82.3	1.5	83.0	-1.1	83.7	-4.6	84.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.8	38.5	42.8	51.9	60.4	69.7	78.0	80.6	80.0	73.7	61.3	48.6	43.2
	DBStd	17.09	11.22	11.14	10.31	8.55	7.26	4.75	4.30	4.74	6.40	8.84	8.82	9.90
	HDD50	1165	387	255	104	17	0	0	0	0	0	12	128	262
	HDD65	3456	822	625	420	185	44	0	0	0	10	178	495	677
	CDD50	5114	30	52	162	328	610	841	949	931	711	363	86	51
	CDD65	1931	0	2	13	46	189	391	484	466	271	65	3	1
	CDH74	20505	0	9	169	611	1927	4169	5251	4966	2619	756	26	2
	CDH80	9721	0	0	34	148	748	2075	2684	2600	1190	242	0	0
Wind	WSAvg	5.7	6.6	7.2	7.1	6.8	5.4	5.0	4.6	4.4	4.5	5.1	5.5	6.2
Precipitation	PrecAvg	50.9	5.0	4.3	5.0	4.4	5.0	3.9	4.4	3.5	3.5	3.2	3.6	5.1
	PrecMax	60.8	9.8	10.0	10.6	7.9	15.6	10.7	8.3	8.5	8.7	8.2	6.2	12.3
	PrecMin	38.5	2.1	2.0	2.8	0.9	0.5	0.1	2.0	1.4	0.2	0.0	0.6	2.4
	PrecStd	5.7	1.6	1.9	1.9	2.0	2.8	2.2	1.7	1.8	2.0	1.9	1.5	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.4	75.3	84.3	88.0	94.0	100.2	102.3	104.4	98.0	91.5	77.6	73.1
		MCWB	60.6	62.7	65.7	67.4	74.3	73.7	76.8	76.1	74.2	68.6	61.2	64.0
	2%	DB	65.6	70.6	79.1	84.4	90.2	96.4	97.8	99.7	93.8	86.1	72.8	68.3
		MCWB	59.2	61.1	63.0	65.9	71.4	75.1	77.2	76.1	73.2	68.6	60.7	61.6
	5%	DB	61.8	66.1	74.1	80.8	87.4	93.5	94.7	95.9	90.9	82.2	68.9	64.2
		MCWB	56.7	58.0	60.2	64.5	70.6	75.1	76.9	76.0	73.0	67.0	59.3	58.7
	10%	DB	57.4	61.8	69.5	77.4	84.4	90.9	92.3	92.4	87.4	78.1	65.0	60.0
		MCWB	51.4	53.7	58.0	63.0	69.6	73.8	76.3	74.7	71.3	65.0	57.2	54.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.7	65.6	67.5	71.4	75.8	79.4	81.0	80.8	77.8	75.0	67.3	66.5
		MCDB	67.6	72.0	79.4	83.4	89.4	92.9	94.6	95.7	88.9	84.8	72.3	70.5
	2%	WB	60.3	63.5	65.0	69.0	74.1	78.2	79.7	78.9	76.5	72.6	64.4	63.1
		MCDB	64.2	68.7	75.9	79.3	86.3	91.3	92.3	92.1	87.4	80.9	68.9	67.0
	5%	WB	57.2	59.9	62.9	66.9	72.8	77.1	78.5	77.7	75.2	70.0	61.6	59.7
		MCDB	61.6	65.0	71.3	76.6	83.8	89.5	90.8	89.5	85.7	77.4	65.8	63.2
	10%	WB	53.0	55.9	60.3	64.7	71.3	75.8	77.6	76.7	73.8	67.4	58.5	55.5
		MCDB	56.4	59.7	67.4	73.8	81.4	86.6	89.4	87.6	82.9	75.1	64.2	59.2
Mean Daily Temperature Range	5% DB	MDBR	19.4	21.0	23.1	25.2	23.0	23.4	22.0	23.0	23.7	25.4	24.0	19.5
		MCDBR	22.9	26.2	29.7	29.4	27.5	27.4	25.6	28.3	28.1	30.0	29.3	23.1
		MCWBR	17.3	17.4	16.2	13.5	11.0	9.0	7.8	8.4	9.7	13.6	17.8	16.8
	5% WB	MCDBR	20.4	21.1	24.7	23.3	22.9	22.8	21.5	22.1	21.9	23.3	19.9	20.0
		MCWBR	17.3	16.3	15.0	12.3	10.2	8.9	7.7	8.1	8.6	12.4	15.1	16.2
Clear-Sky Solar Irradiance	taub	0.320	0.328	0.373	0.402	0.454	0.480	0.513	0.527	0.431	0.377	0.337	0.312	
	taud	2.477	2.447	2.354	2.269	2.169	2.132	2.060	2.006	2.262	2.398	2.463	2.515	
	Ebn at Noon	278	289	284	279	266	258	248	241	261	267	269	274	
	Edh at Noon	27	31	37	42	48	49	53	54	40	32	27	24	
All-Sky Solar Radiation	RadAvg	705	884	1239	1624	1811	1977	1882	1737	1483	1154	832	605	
	RadStd	60	110	116	116	113	148	134	99	84	127	146	87	59

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (22 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	+176	

Nomenclature: See separate page